

Macarthur Diby

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SUMMARY

Software engineer with a B.S. in Computer Science (3.7 GPA) from Texas Tech, currently at Dell Technologies. Builds full-stack and AI/ML systems: LLM distillation pipelines, real-time computer vision, and multi-service cloud deployments. Open to relocation. Fluent in English and French.

PROFESSIONAL EXPERIENCE

Dell Technologies · *Technical Support Engineer* · Austin, TX

Jun 2025 – Present

- Earned Top Performer Gold in two months, ranking in the top tier out of 5,000+ agents company-wide.
- Contributed to AI-driven diagnostic tooling alongside software engineers, supporting testing and implementation cycles that reduced case resolution time by 20%.
- Maintained 98%+ customer satisfaction rate diagnosing complex enterprise hardware and software failures.

WoundCentrics · *Software Engineer* · Lubbock, TX

Jan 2025 – May 2025

- Built Python automation tools for billing workflows and clickstream tracking, cutting processing time 30% and manual work 40%.
- Integrated third-party healthcare APIs and built real-time claims dashboards, reducing rejection rates by 25%.

IT Help Central, Texas Tech University · *IT Systems Technician* · Lubbock, TX

Jul 2022 – Dec 2024

- Resolved 300+ ServiceNow tickets for 150+ students; trained incoming technicians; repeat incidents dropped 15% and resolution time improved 25%.

PROJECTS

LLM Knowledge Distillation Pipeline [\[GitHub\]](#) | *Python, PyTorch, HuggingFace Transformers, Anthropic API, Streamlit*

- Used Claude Haiku to label 1,160 support tickets with soft probability distributions, then fine-tuned DistilBERT as a student model using KL divergence + cross-entropy distillation loss ($T=4$).
- Student runs 65× faster than the teacher at near-zero serving cost, retaining 87% of accuracy across 8 support categories.

Driver Drowsiness Detection System [\[GitHub\]](#) | *Python, OpenCV, dlib, Flask, React Native (Expo), Supabase*

- Real-time drowsiness detection using Eye Aspect Ratio (EAR); three-component architecture: dlib/OpenCV tracker, Flask REST API, and React Native app, with live session updates via Supabase Realtime.
- Three-tier alert system (NORMAL / WARNING / DANGER) driven by event frequency over rolling time windows; no polling on the client side.

AI Debate System [\[GitHub\]](#) | *Node.js, React, OpenAI / Gemini / Groq APIs, AWS EC2 & S3*

- Multi-round debate platform where six AI models argue user-defined topics; separate AI referee models score each round, with the backend on AWS EC2 and frontend on S3.

Patient Appointment System [\[GitHub\]](#) | *Python, Flask, SQLite, JavaScript, HTML/CSS*

- Full-stack scheduling app with user authentication, department routing, and role-based access, built and deployed solo.

Elevator Scheduler [\[GitHub\]](#) | *C++, multithreading, mutexes, condition variables*

- Concurrent elevator dispatcher with thread-safe request queues using POSIX mutexes and condition variables; hybrid algorithm cut average wait time 20–30% vs. distance-only baseline.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C/C++, SQL, HTML/CSS

Frameworks & Libraries: React, Node.js, Flask, React Native, PyTorch, OpenCV, Scikit-learn, HuggingFace Transformers, Streamlit

Cloud & DevOps: AWS (EC2, S3), Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Linux

Databases & Tools: MySQL, PostgreSQL, MongoDB, Supabase | Git, REST APIs, OpenAI API, Gemini API, Anthropic API

EDUCATION

B.S. Computer Science, Minor in Mathematics · Texas Tech University, Lubbock, TX

Dec 2024

GPA: 3.7 | 5× Dean's List | Presidential List | Achiever Scholarship

ACTIVITIES & VOLUNTEER WORK

- NSBE, Texas Tech Chapter Recruiter | Child's Heart, Côte d'Ivoire · Youth volunteer, 50+ children served